

Ha Quoc Viet

DOB: 15/02/2000

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PERSONAL STATEMENT

- Software development with 6 years of experience.
- Backend-focused software engineer with primary expertise in system architecture, API development, and database optimization.
- Expertise in creating user interfaces for web and mobile using ReactJS, Next.js, and React Native.
- Expertise in designing scalable backend solutions using NodeJS, ExpressJS, NestJS, and event-driven architectures with Kafka and Redis.
- Strong foundation in Monolithic and Microservices system design (Event-driven architecture).
- Experienced in database management and optimization with MongoDB, PostgreSQL, and MySQL.
- Worked with both Windows and Unix operating systems.
- Hands-on knowledge in cloud deployment and infrastructure management using AWS (S3, EC2, ECS, CloudFront, RDS, Lambda, EventBridge, Step Functions, SQS).
- English level: B2 to C1 (equivalent to IELTS 6.5).

WORK EXPERIENCE

- **Brighte** Mar. 2025 – Present
Fullstack Developer & Technical Leader at CMC Global
 - Brighte is an Australian fintech company specializing in clean-energy home installations. The platform provides consumer loans and flexible payment plans (including zero-interest options) for sustainable home upgrades such as solar panels, batteries, and energy-efficient systems.
 - Participated in a rigorous interview process with the client, including technical rounds with senior and lead developers, and was successfully selected for the Technical Leader role.
 - Leading a cross-functional team of 12 members to develop and deliver product features with a strong focus on performance, scalability, and timely delivery.
- **Technologies:**
 - Languages: **JavaScript, TypeScript.**
 - Backend: **Node.js, NestJS.**
 - Frontend: **ReactJS, Next.js.**
 - Database: **PostgreSQL, DynamoDB.**
 - Deployment & Cloud: **Terraform, AWS Lambda, AWS EventBridge, AWS Step Functions, AWS SQS, AWS S3, AWS RDS, AWS CloudFront.**

Project: Vendor Webhook

- Designed and developed a comprehensive webhook management module in the Vendor Portal, allowing vendor administrators to create, test, manage, and monitor webhooks.
- Implemented features including webhook creation, editing, deletion, enable/disable toggling, and duplicate prevention based on unique combinations of URL and event.
- Enforced mandatory webhook testing before activation to ensure proper integration and endpoint responsiveness.
- Built retry and locking mechanisms: webhooks are automatically locked after exceeding 5 failed attempts, with email notifications sent to vendor admins.
- Supported integration for 3 core events: **Lead acknowledged, Quote accepted, and Application approved.**
- Developed detailed webhook log views, allowing vendor admins to inspect request/response data and diagnose failures.
- Improved vendor automation workflows and reliability, significantly reducing manual coordination and enhancing real-time integration.

Project: Customer Data Right (CDR) – Income Detection & Calculation

- Designed and implemented a robust income analysis engine using bank transaction data shared via CDR (Consumer Data Right), enabling automated income estimation for credit assessment workflows.
- Developed a multi-phase algorithm combining category-based filtering, clustering, similarity scoring (Levenshtein + amount tolerance), frequency detection, and income normalization.
- Clustered similar transactions using weighted similarity scores (70% description, 30% amount) to identify distinct income streams while mathematically preventing double-counting.
- Applied frequency detection heuristics based on transaction gap analysis to classify income patterns (weekly, fortnightly, monthly, quarterly) and convert them to monthly equivalents.
- Implemented a confidence scoring system factoring in frequency consistency, data sample size, and variance to ensure reliability in financial decision-making.
- Handled edge cases such as base salary vs overtime, bonus detection, seasonal income, and insufficient data using adaptive logic for conservative estimation.
- Produced a comprehensive output including normalized monthly income, inferred frequency, cluster confidence scores, and audit metadata for downstream processing.
- Achieved over 90% accuracy in detecting regular income (e.g. salary/pension), 85%+ frequency classification accuracy, and monthly amount estimations within 5% of actual income.
- Reduced manual income verification workload by 80% and enabled risk-based decision-making through transparent scoring and automated detection.
- Engine designed to handle 95%+ of real-world transaction scenarios with robust quality checks, single-assignment clustering, and progressive tolerance handling.
- **Received high praise and recognition from stakeholders for rapid delivery of the algorithm and initiative in independently designing an effective, production-ready solution.**

Flowinsights

Sep. 2024 – Present

- *Senior Software Engineer at FPT Software*

- Provide Engineering Services to a Norway customer through the Agile Engineering Squads for a privacy-first sensor platform that rapidly converts the physical world into searchable data using fast annotation and powerful processing.

- **Technologies:**

- Languages: **JavaScript, TypeScript.**
- Backend: **Node.js, ExpressJS.**
- Frontend: **ReactJS.**
- Database: **MongoDB, BunnyCDN.**
- Deployment: **Render.**

MXV Exchange Platform

Sep. 2023 – Mar. 2025

- *Fullstack Developer at Mercantile Exchange of Vietnam / Software Development*

- The MXV Exchange Platform is an online trading platform for commodity derivatives in the energy and agricultural sectors. It supports real-time trading and enables domestic investors to access the global CQG platform via the FIX protocol, while also providing access to both real-time and historical market data.
 - * Designed and implemented a microservices-based architecture to handle high-volume trading transactions.
 - * Developed scalable backend APIs with ExpressJS and NestJS, ensuring secure and high-performance order execution.
 - * Enabled vertical scaling by utilizing worker threads and Node.js built-in cluster to fully utilize CPU resources.
 - * Integrated Kafka and Redis for real-time data processing and event-driven services.
 - * Optimized database schema design for MongoDB and PostgreSQL to improve query performance.
 - * Optimized database performance through efficient database configuration, connection pooling, SQL query tuning, and the implementation of master-slave architecture, partitioning, sharding, and other performance-enhancing techniques.
 - * Managed Dockerized deployments with Nginx for scalable backend services.
 - * Built the frontend in Next.js and React Native based on user requirements and Figma, integrating efficiently with the backend system.
- **Technologies:**
 - * Languages: **JavaScript, TypeScript.**

- * Backend: **NodeJS, ExpressJS, NestJS, Kafka, Redis.**
- * Frontend: **Next.js.**
- * Database: **MongoDB, PostgreSQL.**
- * Deployment: **Docker, Nginx.**

- VitaDairy & Traphaco ZMA (Second Job as a Freelancer)** May. 2023 – Jun. 2025
 - Freelance Developer at Tesosoft / Software Development*
 - Developed and optimized UI for VitaDairy's admin management.
 - Developed and optimized UI for Traphaco Zalo mini app.
 - Refactored backend logic for Traphaco ZMA and Vitadairy loyalty application, fixing critical bugs.
 - Technologies:**
 - * Languages: **JavaScript, TypeScript.**
 - * Backend: **NodeJS, NestJS, Redis.**
 - * Frontend: **React.js.**
 - * Database: **PostgreSQL.**
 - * Deployment & Cloud services: **AWS S3, AWS RDS.**
- Amai Content** Dec. 2021 – Sep. 2023
 - Backend Developer at Amai Agency / Software Development*
 - Amai Content is an application designed to optimize content and manage multi-platform social networks such as Facebook, Google My Business, Instagram, and more.
 - * Built a transaction management system for automated payments integrated with Vietcombank and A Chau Bank.
 - * Developed a real-time messaging system for multi-platform content management using Facebook Messenger API.
 - * Created backend services using Google Business Profile API, enabling data synchronization, analytics, and media management.
 - * Developed an automated workflow for publishing content across multiple social media platforms.
 - * Designed efficient MongoDB and MySQL schemas to optimize database queries.
 - Technologies:**
 - * Languages: **JavaScript, TypeScript.**
 - * Backend: **NodeJS, ExpressJS.**
 - * Database: **MySQL, MongoDB.**
- JUSEI Master** Feb. 2021 – Dec. 2021
 - Backend Developer at Chainos Solution / Software Development*
 - JUSEI Master is a study application for the national examination for Judo therapy experts.
 - * Developed data analysis and statistics features, optimizing performance of backend queries.
 - * Designed and implemented scalable MongoDB schemas to handle large data sets efficiently.
 - Technologies:**
 - * Languages: **TypeScript.**
 - * Backend: **NestJS.**
 - * Database: **MongoDB.**
- Re.Monster** Mar. 2020 – Jan. 2021
 - Fresher Frontend Developer at Chainos Solution / Software Development*
 - In the world of Re.Monster, trainers take center stage, raising monsters and competing against one another. Monsters are regenerated based on on-chain data and can be obtained by using NFTs stored in the player's wallet.
 - Link: <https://remonster.world/>
 - Technologies: TypeScript, React.js**
- Kanimal Clash** Mar. 2020 – Jan. 2021
 - Fresher Frontend Developer at Chainos Solution / Software Development*
 - Kanimal Clash is a real-time P2E (Play-to-Earn) strategy game developed and published by Ludena Protocol. Players can collect and breed 32 types of unique Kanimal NFTs to win PvP battles for incredible rewards.
 - Link: <https://ludena-protocol.gitbook.io/kanimal-clash/kanimal-clash>
 - Technologies: TypeScript, React.js**

ADDITIONAL SKILLS

- English Level: **B2**
- Version Control: **Git, GitHub, GitLab.**
- Development Tools: **Visual Studio Code, Jira, Trello.**
- Automation & Scraping: **Puppeteer, Selenium.**

EDUCATION

- University: **Vietnam National University – Ha Noi University of Science**
- Major: **Computer & Information Science Advanced Program.**